

Case Study

Retail: Store Sales Performance Analysis

Problem Statement

The regional sales leadership team wants to understand which regions, product categories, and discount strategies are actually driving profitable growth—not just generating revenue. While sales have increased over time, management suspects that aggressive discounting is eroding profit margins, particularly in certain product categories and sub-categories.

Without a clear understanding of profitability drivers, the company risks investing in products and regions that appear successful based on sales alone but contribute little—or even negatively—to overall business performance.

The objective of this analysis is to identify the most profitable products, regions, customer segments, and discount strategies while providing actionable recommendations to improve profitability and optimize future sales decisions.

Data Overview

- **Source:** Global Superstore Retail Dataset
- **Records:** ~10,000 Orders
- **Tools Used:** Excel, MySQL, Power BI

The dataset contains order information, shipping details, customer segments, geographic regions, product categories, sales, quantity, discounts, profit, profit margins, and time-based sales attributes.

Tool Stack

Excel (Data Cleaning) → MySQL (Business Analysis) → Power BI (Interactive Dashboard)

Business Questions

1. Which region generates the highest sales and the highest profit?
2. Which product categories deliver the strongest profit margins?
3. Which sub-categories are generating losses despite high sales?
4. How do discounts affect profitability?
5. Are there seasonal sales trends throughout the year?
6. Which customer segments contribute the highest revenue and profit?
7. Which products should receive greater marketing investment?

Data Cleaning (Excel)

Performed data validation and quality checks before analysis.

Cleaning Steps

- Verified there were no duplicate order records.
- Checked for missing or blank values.
- Standardized text formatting across categorical fields.
- Converted Order Date and Ship Date into proper date format.
- Verified Sales, Profit, Quantity, and Discount values.
- Created Profit Margin (%) column.
- Extracted:
 - Order Year
 - Order Quarter
- Created Discount Bands:
 - 0%
 - 1–20%
 - 21–40%
 - Above 40%

Analysis (SQL)

Key Findings

1. Which region has the highest and lowest total sales?

Although **West** generated the highest total sales (\$**725457.93**), and delivered the strongest profit margin (**14.94%**).

Other side **South** generated the lowest total sales demonstrating that revenue alone is not an accurate measure of business performance.

Region	Total_Order	Total_Sales	Total_profits	Profit_Margin_Pct	Total_Quantity
Central	1175	501239.88	39706.45	7.92	8780
East	1401	678781.36	91522.84	13.48	10618
South	822	391721.90	46749.71	11.93	6209
West	1611	725457.93	108418.79	14.94	12266

2. Which product categories are most profitable?

The **Technology** category recorded the highest average profit margin (**17.40%**) despite representing a smaller share of total orders of **1544**.

Meanwhile, the **Furniture** category produced the weakest margins of **2.49%**, suggesting pricing pressure and excessive discounting.

Category	Total_Order	Total_Sales	Total_profits	Profit_Margin_Pct	Total_Quantity
Furniture	1764	741999.98	18451.25	2.49	8028
Office Supplies	3742	719046.99	122490.88	17.04	22906
Technology	1544	836154.10	145455.66	17.40	6939

3. Are any product categories selling well but losing money (negative margin)?

The biggest profitability concern was **Tables** (Sub-Category of Furniture), which generated approximately **\$206965.68** in sales while producing **-17725.59** in losses.

Increasing sales volume within this category actually reduces overall company profitability.

Category	Sub_category	Total_Order	Total_Sales	Total_profits	Profit_Margin_Pct	Total_Quantity
Furniture	Tables	307	206965.68	-17725.59	-8.56	1241
Furniture	Bookcases	224	114880.05	-3472.56	-3.02	868
Office Supplies	Supplies	187	46673.52	-1188.99	-2.55	647
Technology	Machines	112	189238.68	3384.73	1.79	440
Furniture	Chairs	576	328449.13	26590.15	8.10	2356
Office Supplies	Storage	777	223843.59	21279.05	9.51	3158
Technology	Phones	814	330007.10	44516.25	13.49	3289
Furniture	Furnishings	877	91705.12	13059.25	14.24	3563
Office Supplies	Binders	1316	203412.77	30221.64	14.86	5974
Office Supplies	Appliances	451	107532.14	18138.07	16.87	1729
Office Supplies	Art	731	27118.80	6527.96	24.07	3000
Technology	Accessories	718	167380.31	41936.78	25.05	2976
Office Supplies	Fasteners	215	3024.25	949.53	31.40	914
Technology	Copiers	68	149528.01	55617.90	37.20	234
Office Supplies	Envelopes	249	16476.38	6964.10	42.27	906
Office Supplies	Paper	1191	78479.24	34053.34	43.39	5178
Office Supplies	Labels	346	12486.30	5546.18	44.42	1400

4. Does discounting improve business performance?

Higher discounts consistently reduced profitability.

Orders receiving discounts above **40%** generated an average profit of **\$-106.71**, indicating that excessive discounting destroys margin rather than stimulating profitable growth.

Discount_Band	total_orders	avg_order_sales	avg_quantity	total_profit	avg_profit_per_order	profit_margin_pct
1. No Discount	2644	226.74	3.8	320987.88	66.90	29.51
2. Low (1-20%)	2507	222.59	3.7	100786.35	26.50	11.91
3. Mid (21-40%)	400	509.00	3.8	-35817.58	-77.86	-15.30
4. High (41%+)	737	137.87	3.9	-99558.86	-106.71	-77.40

5. Is there seasonality in sales?

Sales consistently peak during **Q4 (October–December)** across all years.

Holiday demand represents the company's strongest revenue opportunity and requires proactive inventory planning.

Order_Year	Order_Quarter	Year_Quarter	Total_Orders	Total_Sales	Total_Profit
2014	1	2014-Q1	131	74447.86	3811.20
2014	2	2014-Q2	201	86538.77	11204.16
2014	3	2014-Q3	267	143633.18	12804.73
2014	4	2014-Q4	370	179627.75	21723.97
2015	1	2015-Q1	144	68851.74	9264.94
2015	2	2015-Q2	214	89124.28	12190.92
2015	3	2015-Q3	274	130259.51	16853.70
2015	4	2015-Q4	406	182296.93	23309.13
2016	1	2016-Q1	179	93237.20	11441.39
2016	2	2016-Q2	294	136082.33	16390.30
2016	3	2016-Q3	378	143787.43	15823.63
2016	4	2016-Q4	464	236098.90	38139.95
2017	1	2017-Q1	240	123144.84	23506.21
2017	2	2017-Q2	367	133764.33	15499.42
2017	3	2017-Q3	448	196251.94	26985.26
2017	4	2017-Q4	632	280054.08	27448.88

6. Which sub-categories receive the highest discounts?

Binders receives the highest average discount (**37.2%**) while simultaneously recording one of the weakest profit margin.

This suggests that discount strategies are compensating for weak pricing rather than creating sustainable demand.

category	Sub_category	Total_Order	Avg_Discount	Total_Profit	Total_Sales
Office Supplies	Binders	1316	37.2	30221.64	203412.77
Technology	Machines	112	30.6	3384.73	189238.68
Furniture	Tables	307	26.1	-17725.59	206965.68
Furniture	Bookcases	224	21.1	-3472.56	114880.05
Furniture	Chairs	576	17.0	26590.15	328449.13
Office Supplies	Appliances	451	16.7	18138.07	107532.14
Technology	Copiers	68	16.2	55617.90	149528.01
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Office Supplies	Paper	1191	7.5	34053.34	78479.24
Office Supplies	Storage	777	7.5	21279.05	223843.59
Office Supplies	Labels	346	6.9	5546.18	12486.30

Other Findings

1. Customer Segment Performance

The **Consumer** customer segment contributes the highest overall sales and profit, making it the strongest target for future marketing campaigns and customer retention initiatives.

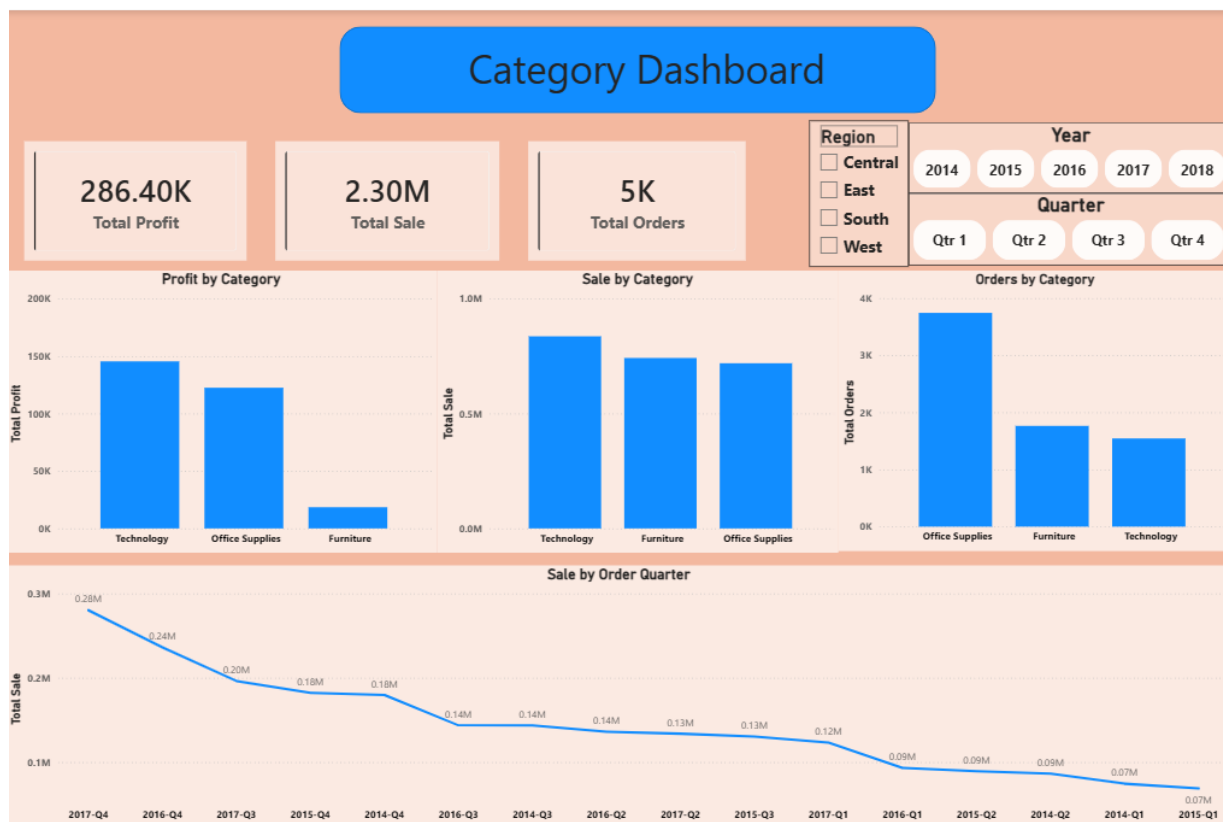
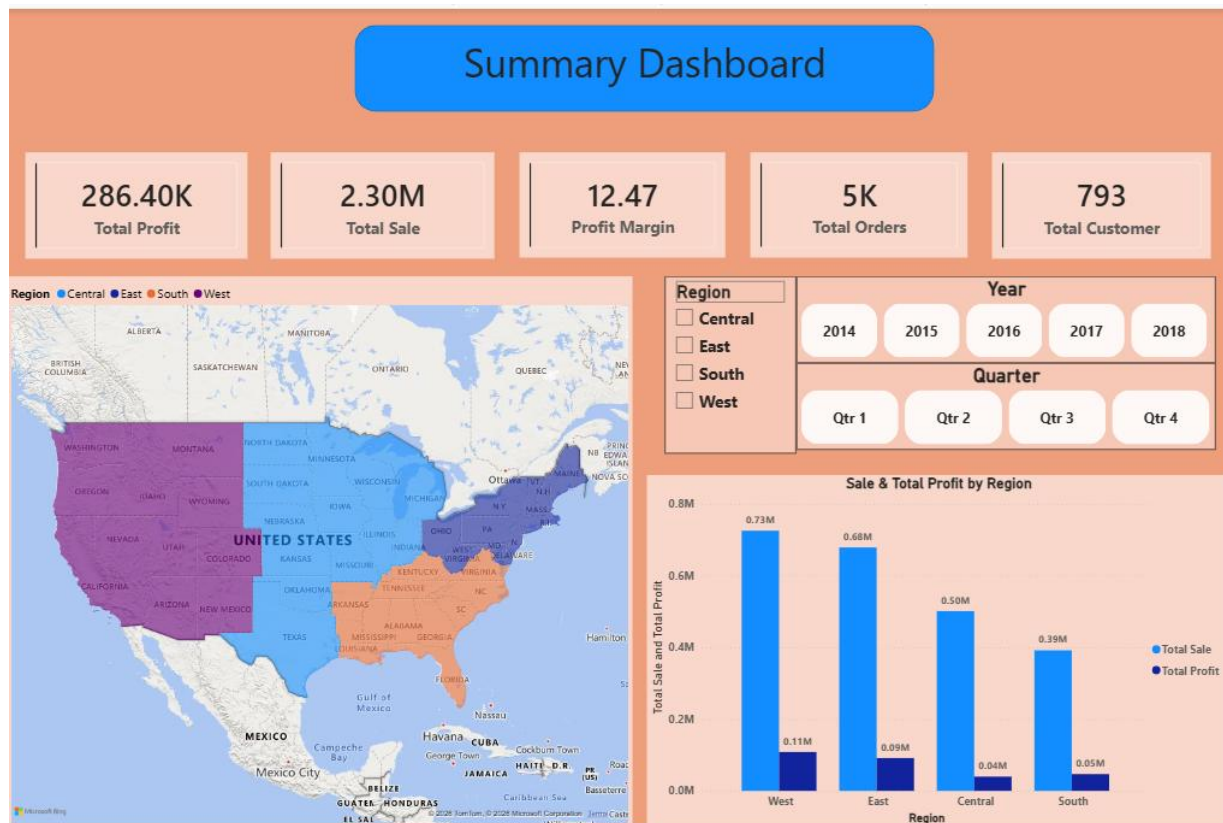
segment	Total_Order	Total_Sales	Total_profits	Profit_Margin_Pct	Total_Quantity
Consumer	2586	1161401.34	134119.33	11.55	19521
Corporate	1514	706146.44	91979.45	13.03	11608
Home Office	909	429653.29	60299.01	14.03	6744

2. Customer Segment Performance

The **Standard Class** shipping Mode contributes the highest overall (**\$ 1358216.08**) sales and (**\$ 164089.25**) profit, making it the best shipping mode, other hand **Same Day** has lower sales and profits.

ship_Mode	Total_Order	Total_Sales	Total_profits	Profit_Margin_Pct	Total_Quantity
First Class	787	351428.43	48969.95	13.93	5693
Same Day	264	128363.12	15891.90	12.38	1960
Second Class	964	459193.44	57446.49	12.51	7423
Standard Class	2994	1358216.08	164089.45	12.08	22797

Dashboard (Power BI)



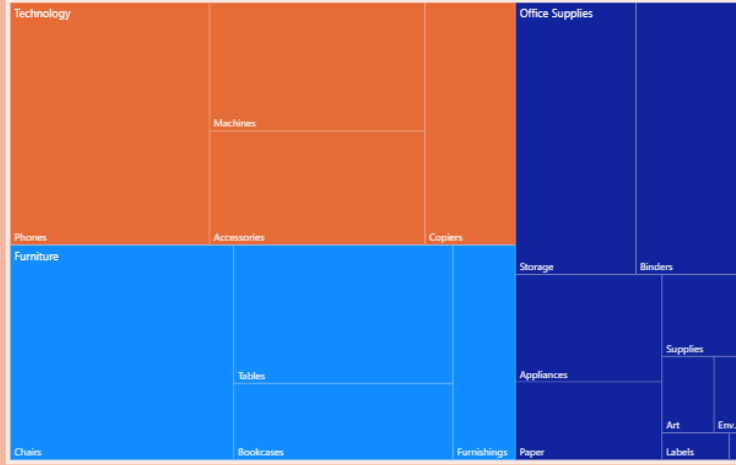
Sub-Category Dashboard

286.40K
Total Profit

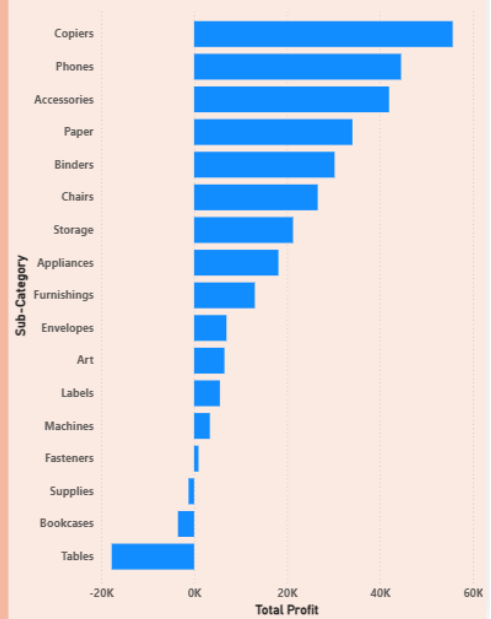
2.30M
Total Sale

5K
Total Orders

Sale by Category and Sub-Category



Profit by Sub-Category



Shipping Dashboard

286.40K
Total Profit

2.30M
Total Sale

5K
Total Orders

Region
☐ Central
☐ East
☐ South
☐ West

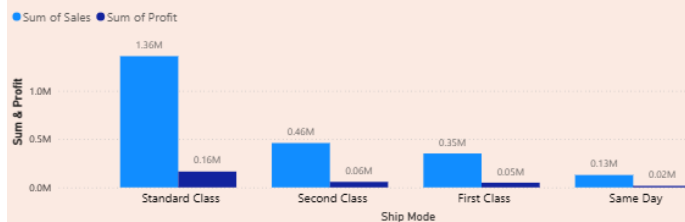
Year

2014 2015 2016 2017 2018

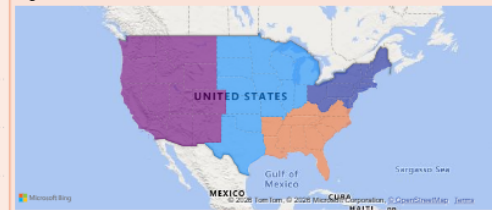
Quarter

Qtr 1 Qtr 2 Qtr 3 Qtr 4

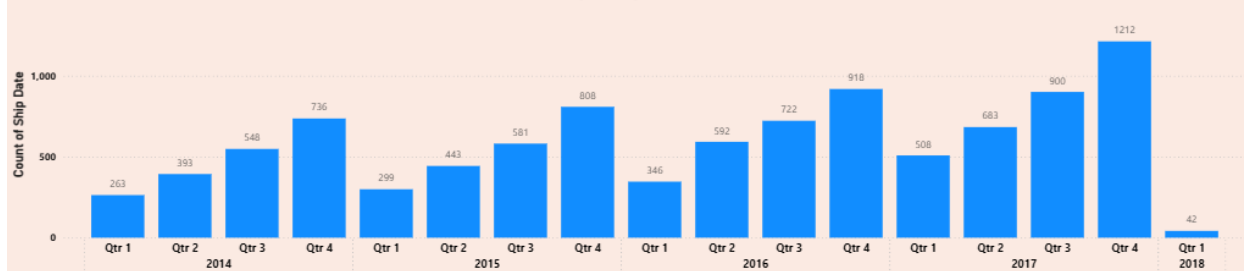
Sales and Profit by Ship Mode



Region ● Central ● East ● South ● West



Count of Ship Date by Year and Quarter



Recommendations

1. Reduce excessive discounting

Immediately review discount policies for loss-making sub-categories.

Products consistently sold at discounts above **40%** should require management approval before promotion.

2. Reevaluate Furniture pricing

Furniture generates strong revenue but weak profitability.

Review supplier costs, pricing strategy, and discount policies to improve margins.

3. Invest in Technology products

Technology consistently delivers the strongest profit margins.

Increase inventory availability and marketing investment for high-margin products.

4. Prepare inventory ahead of Q4

Historical sales patterns show that Q4 generates the highest demand.

Forecast inventory requirements at least three months in advance to avoid stock shortages.

5. Introduce profit-based performance metrics

Evaluate business performance using both **Sales** and **Profit Margin** instead of revenue alone to prevent growth driven by unprofitable sales.

6. Monitor high-risk sub-categories

Create monthly profitability reports to identify products that generate high sales but consistently lose money.

7. Implement discount governance

Require approval for discounts above **20%**, supported by a projected margin impact analysis before implementation.

